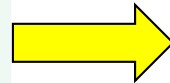


Why Assignments Are Too Easy?

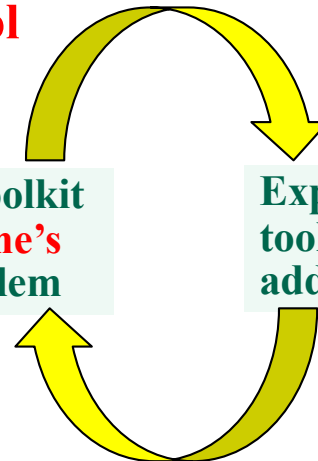
A: It's intentional. To provide hands-on seeds, so as to start using tools (MPI, OpenMP, CUDA, SYCL, *etc.*, or other mathematical & engineering tools at large) in a self-sustained use-learn cycle

Most effective way to learn a new tool

Understand
the smallest set
of essential
mechanisms
by reading a
**minimalist
program**

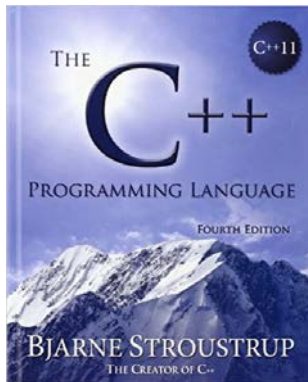


Use the toolkit
to solve **one's
own** problem



Expand the
toolkit by
adding

**Self-sustained
Use-learn cycle**



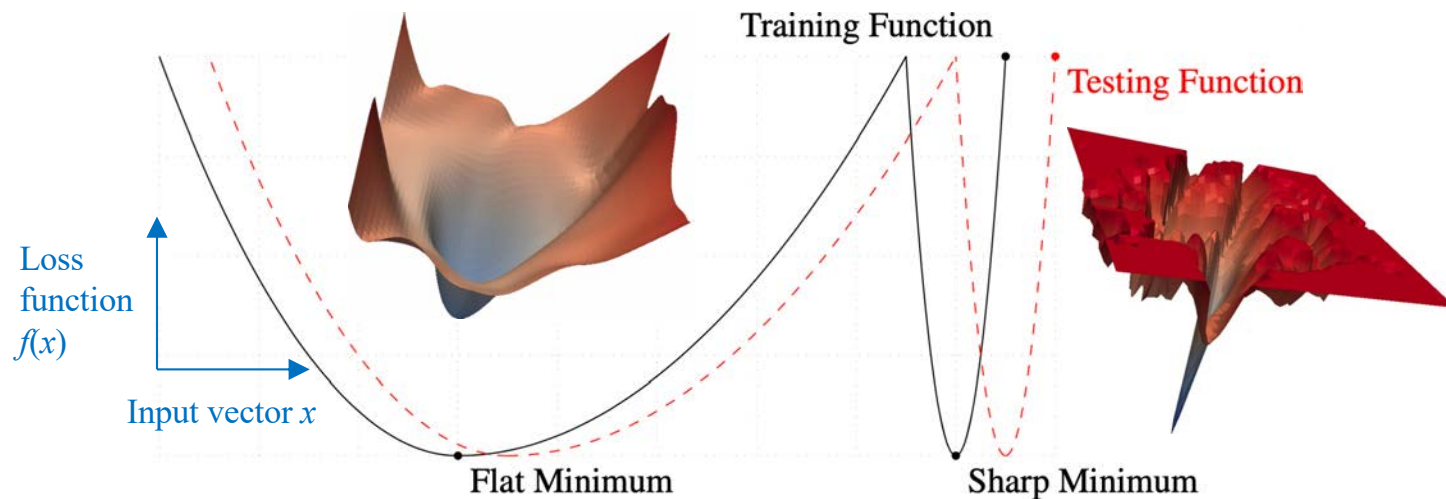
Start the cycle in your final project

Alternative way

“You bought a 1300+ page (480K+ word) book for \$40 or \$60, not a subscription service. How much support do you expect?”

Learning-Theory Foundation

- **Minimum description length (MDL) theory:** Models described by fewer bits (*i.e.*, low Kolmogorov complexity) are more **generalizable**
- **Spectral theory:** Low-bit (low-complexity) models have **flatter loss landscape**



Keskar et al., *ICLR 2017* [<https://arxiv.org/abs/1609.04836>]; Forest et al., *ICLR 2021* [<https://arxiv.org/abs/2010.01412>]

H. Ibayashi et al., *ISC23—LNCS 13948*, 223 ('23); *arXiv*: 2303.08169



Allegro-Legato

<https://github.com/mir-group/allegro>

- Also important is **convexity** in geometric cognitive space in cognitive science

Tetkova et al., *Nat. Commun.* **16**, 5419 ('25)

Put Hands On, First Time

- Do it yourself to make knowledge “grounded”; once operationally understood, you can ask AI to repeat it as many times needed *Least time-consuming assignments
We will do it together in the class*
- Get sensory grounding (*i.e.*, connect abstract symbol to physical experience)
cf. Helen Keller’s water moment <https://www.youtube.com/watch?v=LTsRVYq9J0Q>
T. Zahavy, ICML ('26) <https://openreview.net/forum?id=klU4737opt>
- Or else ...
You will get “a smoother version of own misunderstanding,” *i.e.*,
“LLM is an ignorance multiplier” (Tobias Osborne)
- Would AI agents living in world model experience water moment?

LLMs can't jump!



or perhaps can

